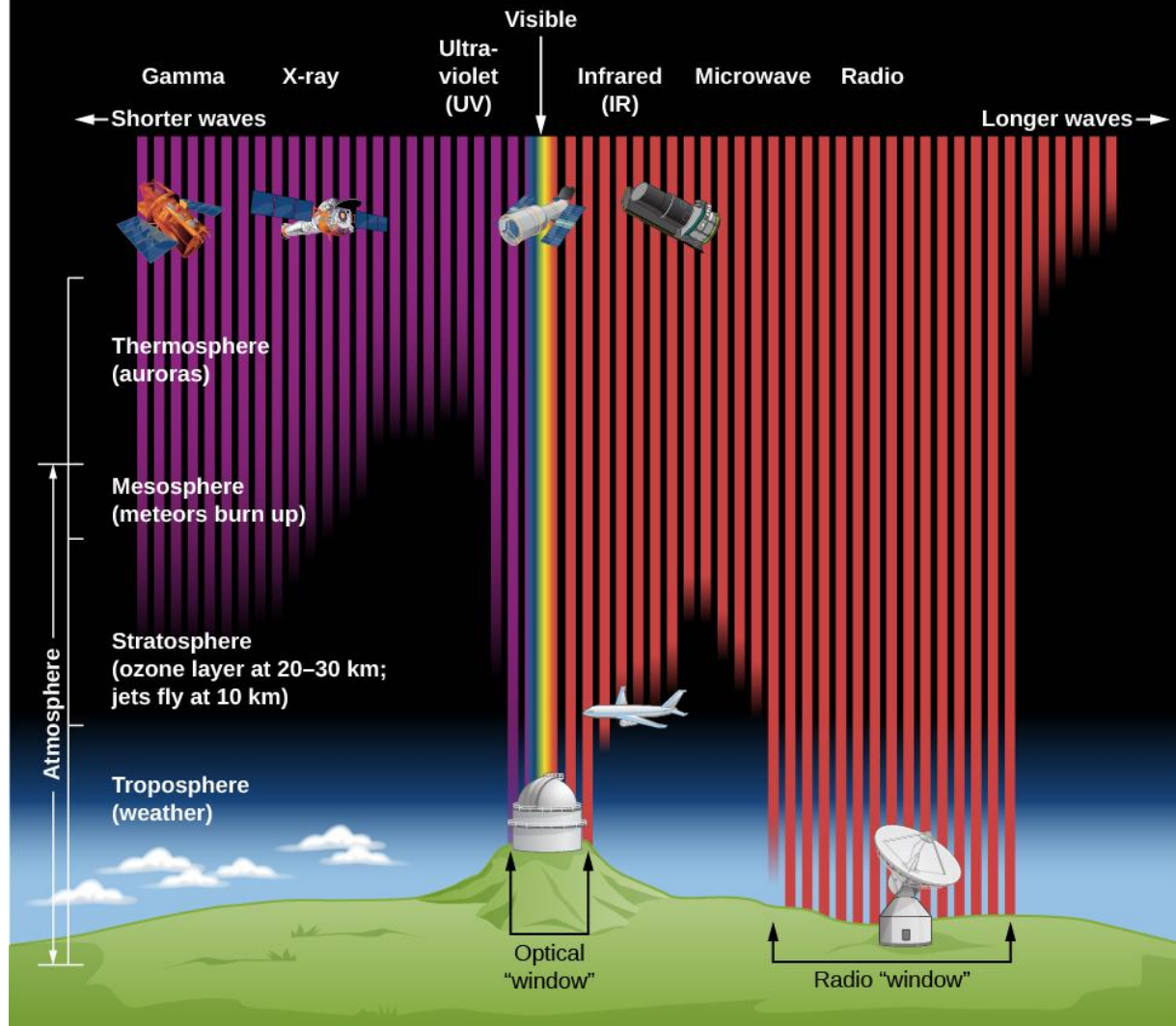


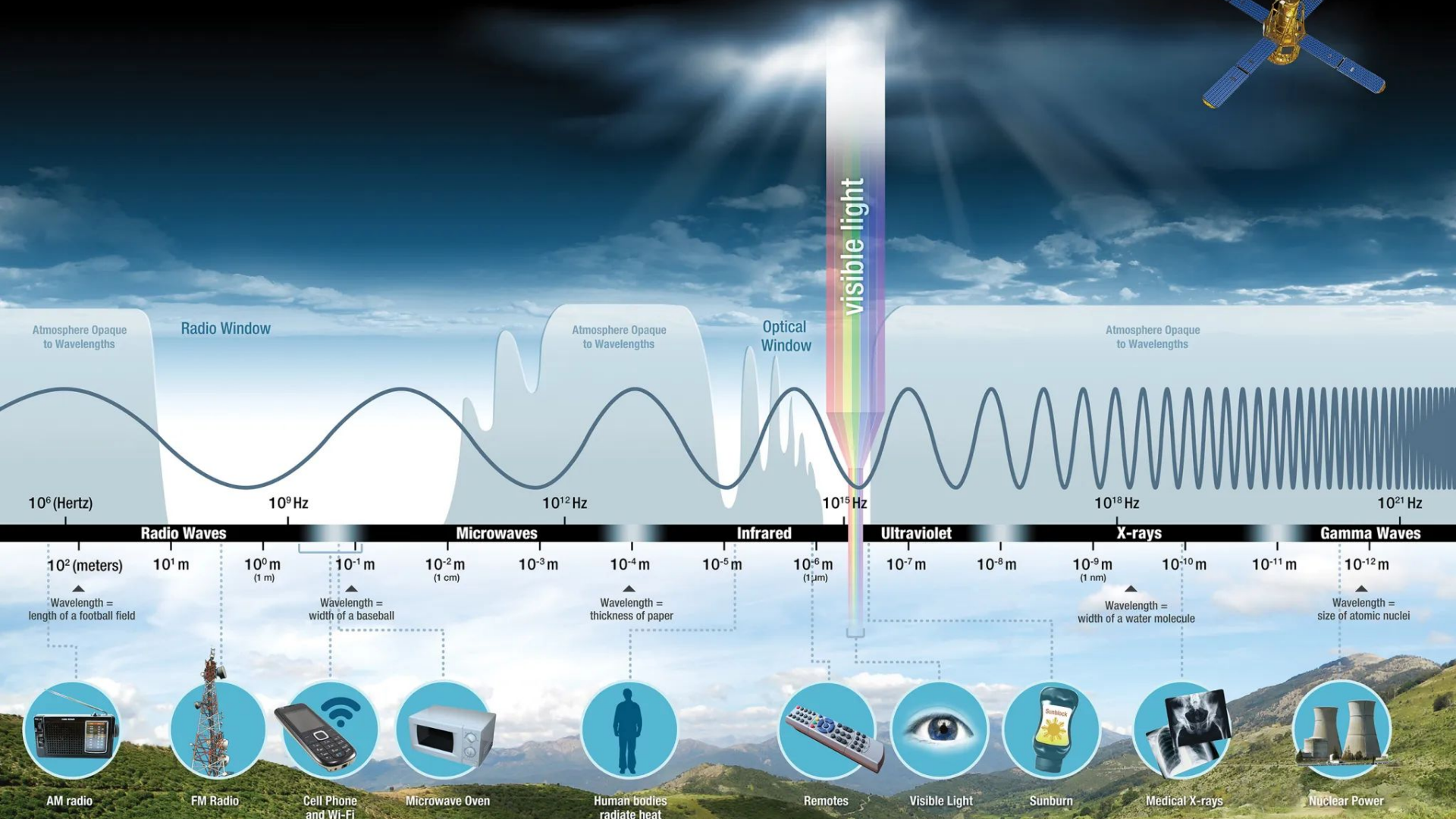
Astro 465

Optical part

Rough plan

1. Intro to optical observations with CCDs
 - a. telescope types & components, observing lingo
 - b. CCD function and instrumental artifacts
 - c. Considerations for observing, finder & airmass charts
2. From raw data to science
 - a. data reduction: bias, darks, flatfields
 - b. magnitude systems
 - c. calibrations: photometry & astrometry
3. Finding sources, photometry
 - a. peak detection vs filtering
 - b. aperture, PSF, and differential photometry
4. Catalog matching, spectral energy distributions, color images
5. Strategies for good data stewardship





Preview to Optical Observing Labs

Lab 1: CCD Basics – characterizing the CCD detector, preparing for real data
1 week

Lab 2: Star cluster Photometry & isochrone fitting
2 weeks

Lab 3: Galaxy surface photometry & disk/bulge decomposition
2 weeks

Main challenges: Prepare a target list for both
Find a clear night for actually taking data

Star Clusters

Why study star clusters:

Virtually all stars are born in clusters

All cluster members were born from same gas:

→ Same age, same metallicity

Star Clusters are great labs for:

- Studies of stellar evolution
- Galaxy Structure
- Formation and Evolution of galaxies both in our Milky Way and even in more distant galaxies

Open
Cluster



Globular
Cluster



Star Clusters

Why study star clusters:

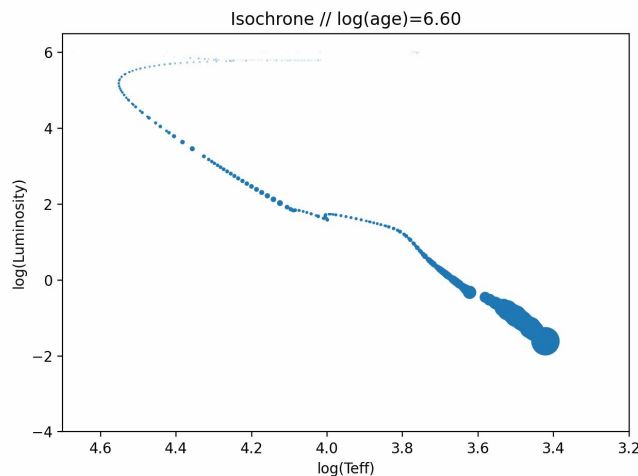
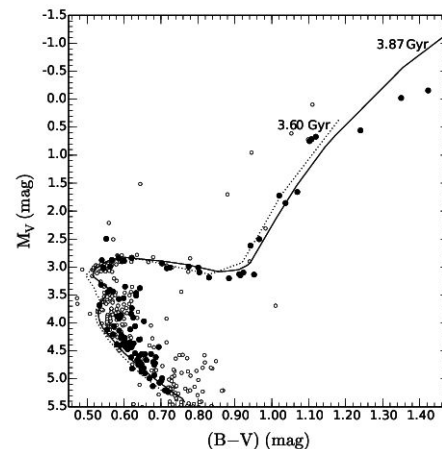
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Star Clusters are great labs for:

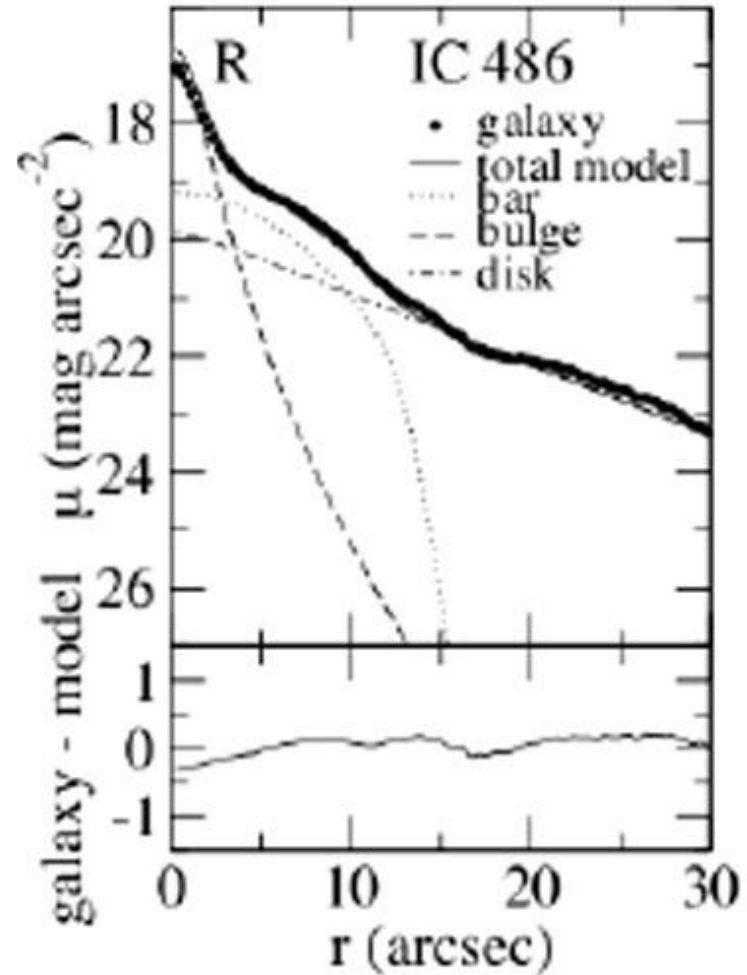
- Studies of stellar evolution
- Galaxy Structure
- Formation and Evolution of galaxies
both in our Milky Way and even in
more distant galaxies



Galaxy Surface Photometry

Beyond integrated photometry of galaxies, surface brightness photometry can reveal:

- Structure of galaxies: Sizes, shapes, distribution of stars
- Insights into stellar populations
→ formation history
- Commonalities and difference between galaxies



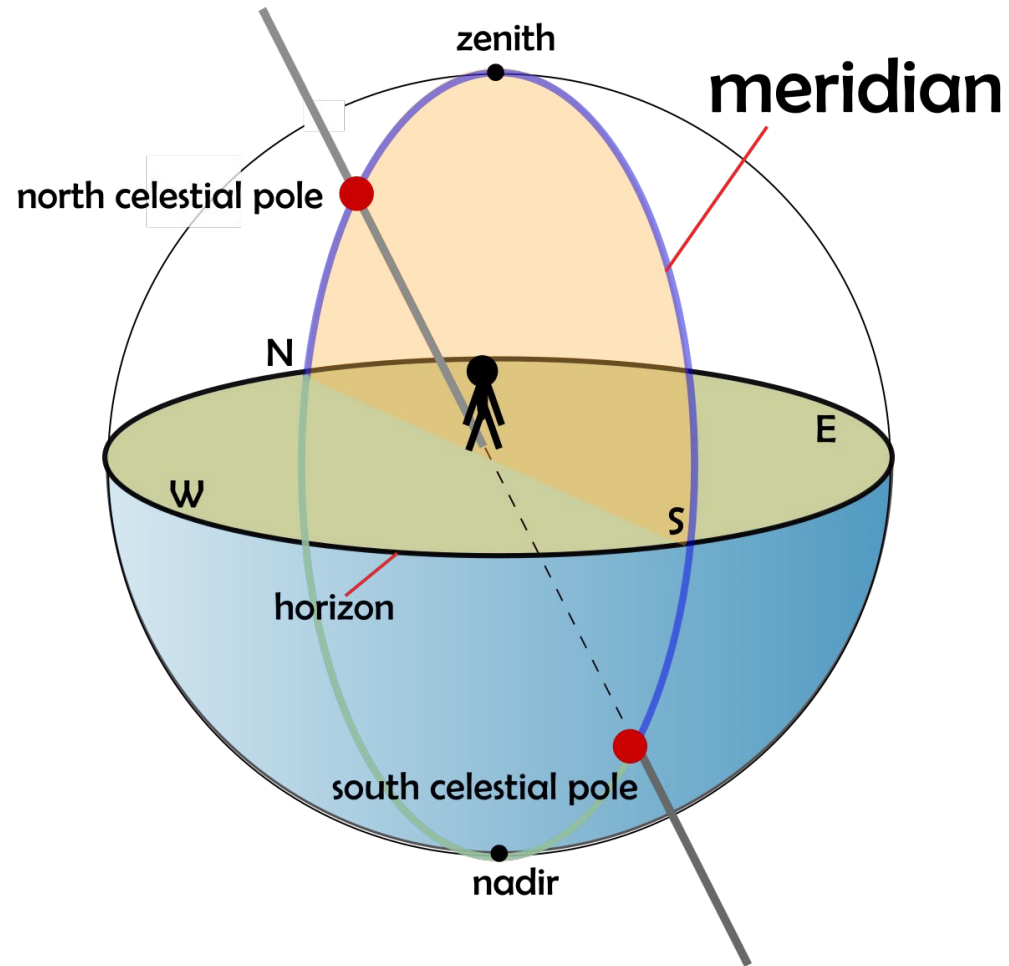
Preparations

- Select targets for both Labs (start with 1 star cluster, but have 2nd ready) – see lab instructions for details on target catalog
 - If in doubt, email me to make sure it's suitable
- Check observability during the next 2-3 weeks
- Once we have a favorable weather forecast: Sign-up sheet on Google Docs
- Take calibrations before/around sunset
- To make sure everyone gets a chance to take some:
 - 1 target per observer for each lab, then exchange/share observations.
 - Once everyone has data we can take some more.

Astronomer Speak 101

Important points in the sky:

- Zenith: Point right overhead
- Meridian: Great circle passing through zenith and due south



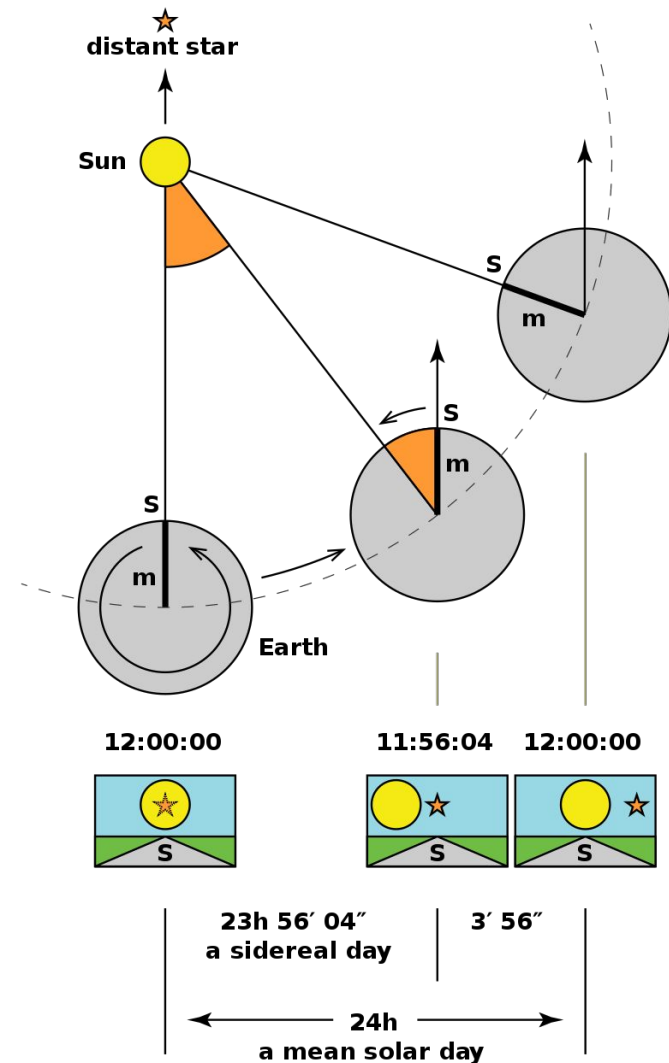
Astronomer Speak 101: Sidereal time

The astronomical (sidereal) day is a bit shorter than the solar (regular) day.

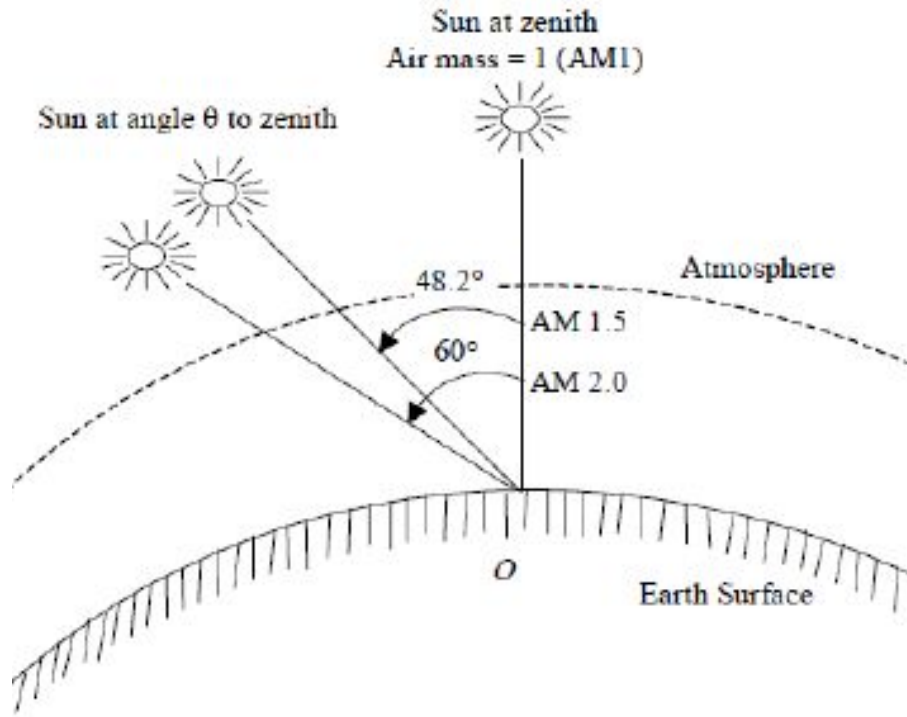
At the vernal equinox (i.e. September 23rd), sidereal time and solar time coincide.

In practice, the local apparent sidereal time (LAST) indicates what right ascension is crossing the meridian at this moment. This is also the moment when objects are highest in the sky.

You can look up the current local sidereal time at <https://www.localsiderealtime.com/>



Astronomer Speak 101: Airmass

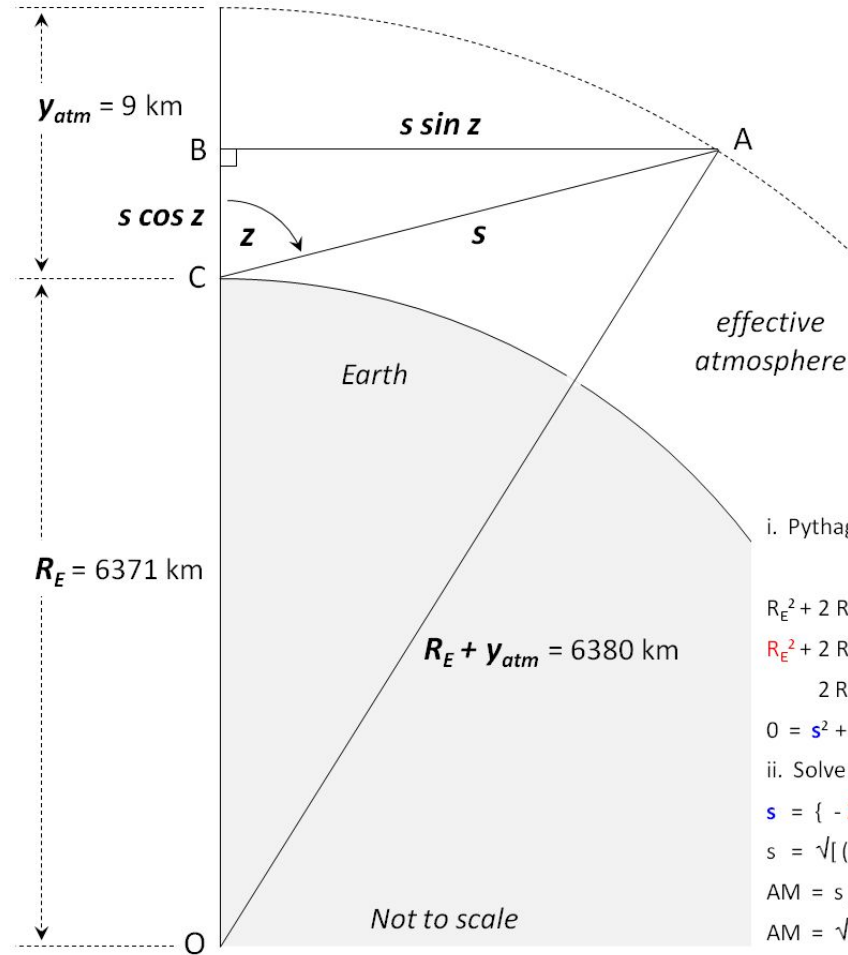


Airmass is a way of characterizing through how much air we are observing an astronomical source.

Higher airmasses (i.e. larger zenith distances) mean

- more extinction along the way
- more turbulence (reduced image quality)

When planning what to observe when, try to minimize the airmass at the time of observation.



i. Pythagoras applied to right-angle triangle OAB :

$$(R_E + y_{atm})^2 = (R_E + s \cos z)^2 + (s \sin z)^2$$

$$R_E^2 + 2 R_E y_{atm} + y_{atm}^2 = R_E^2 + 2 R_E s \cos z + s^2 \cos^2 z + s^2 \sin^2 z$$

$$R_E^2 + 2 R_E y_{atm} + y_{atm}^2 = R_E^2 + 2 R_E s \cos z + s^2 (\cos^2 z + \sin^2 z)$$

$$2 R_E y_{atm} + y_{atm}^2 = 2 R_E s \cos z + s^2$$

$$0 = s^2 + 2 R_E s \cos z - (2 R_E y_{atm} + y_{atm}^2)$$

ii. Solve quadratic for s :

$$s = \left\{ -2 R_E \cos z \pm \sqrt{(2 R_E \cos z)^2 + 4(2 R_E y_{atm} + y_{atm}^2)} \right\} / 2$$

$$s = \sqrt{(R_E \cos z)^2 + 2 R_E y_{atm} + y_{atm}^2} - R_E \cos z$$

$$AM = s / y_{atm} ; r = R_E / y_{atm} \approx 708$$

$$AM = \sqrt{[r^2 \cos^2 z + 2r + 1]} - r \cos z$$

Best practices for observing my target

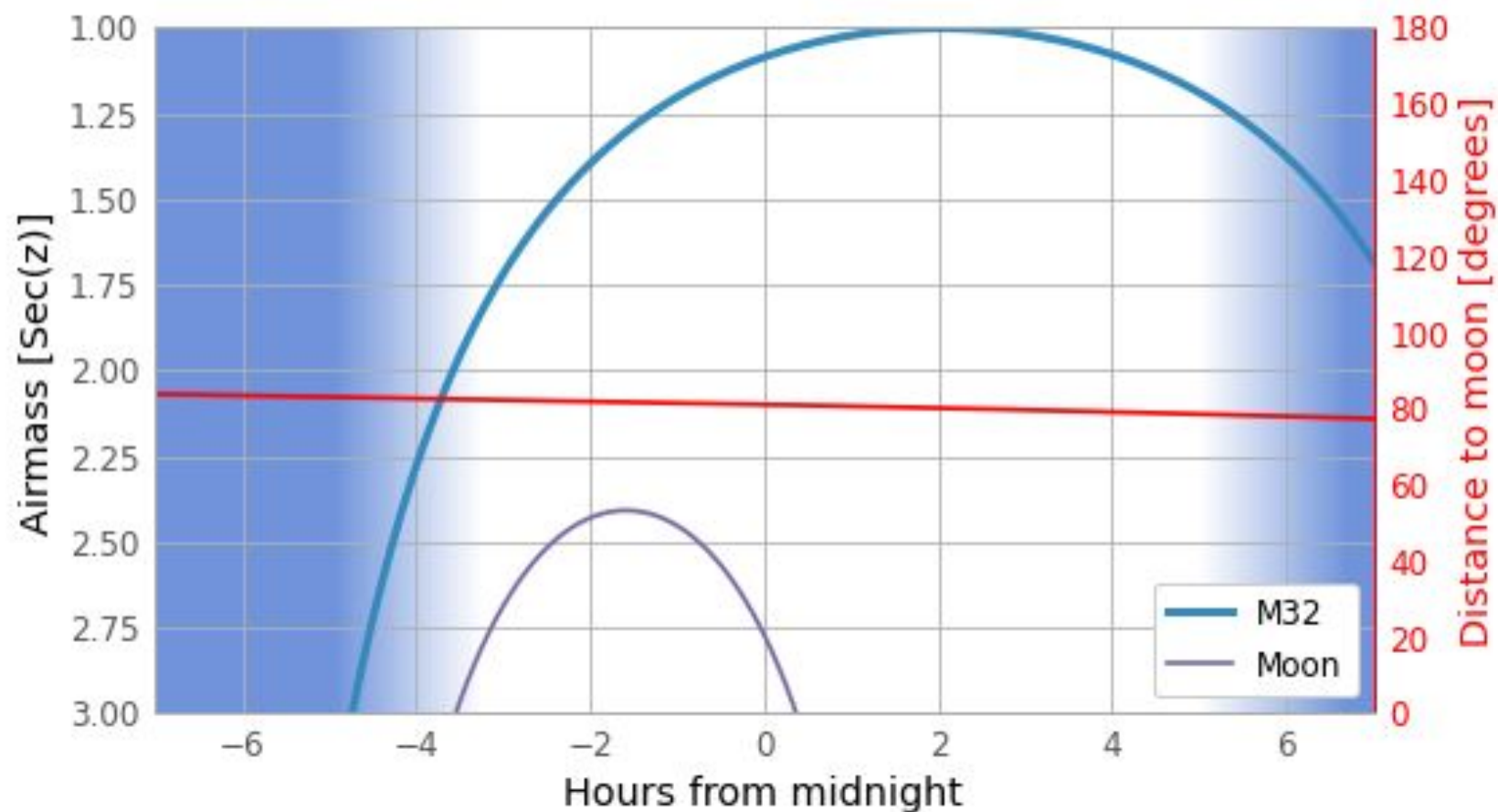
let's do some brainstorming ...

Best practices for observing my target

- Clear skies !!!
- Dark sky → little moon light, so around new moon (less critical in Madison, since sky is bright anyway)
- If there is moon: Further away from moon is better (60+ degrees is good)
- Minimal airmass, so object high in the sky
 - declination ~ latitude of observer → ~40 degrees (10-70 degrees is good)
 - object near meridian (due south) → RA = LAST
- Filter choice:
 - wider wavelength coverage is better, but:
 - bluer bands have more scattered lights from Madison
 - bluest & reddest bands have too low detector sensitivity
- Exposure times: Long enough but not too long
- Binning: Aim for binned pixelscale ~ 0.5x seeing → likely 2x2 or 3x3

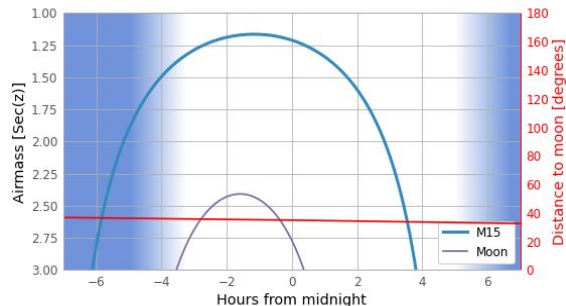
Visibility Chart for M32 on Sun, September 15, 2024

RA: 0h42m41.8248s DEC: 40d51m54.612s

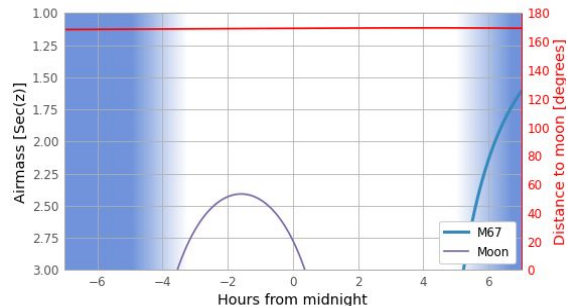


Selecting suitable targets

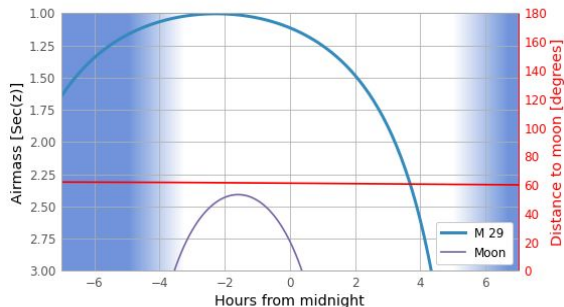
Visibility Chart for M15 on Sun, September 15, 2024
RA: 21h29m58.33008s DEC: 12d10m01.2s



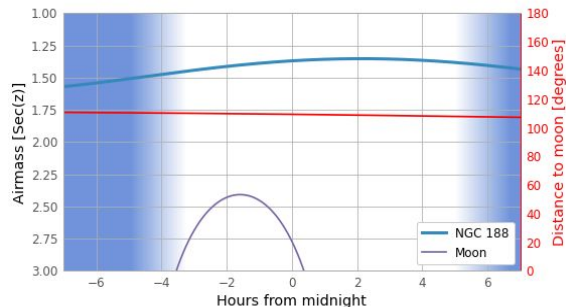
Visibility Chart for M67 on Sun, September 15, 2024
RA: 8h51m23.04s DEC: 11d48m50.4s



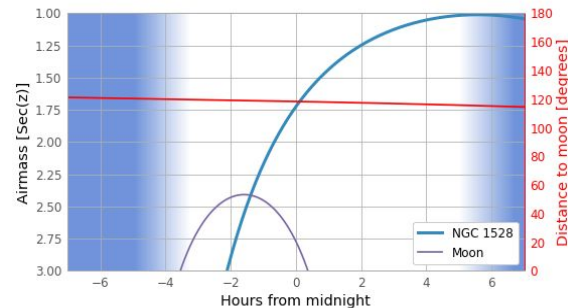
Visibility Chart for M 29 on Sun, September 15, 2024
RA: 20h23m46.32s DEC: 38d29m13.2s



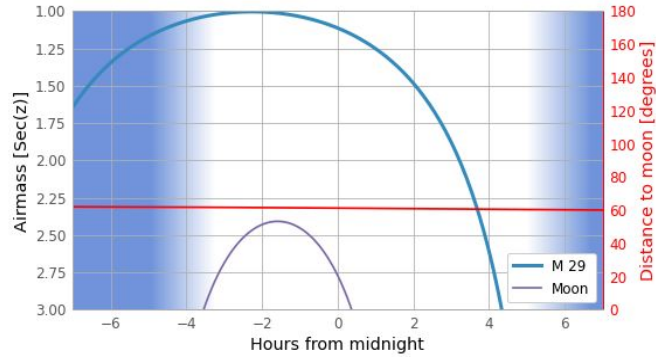
Visibility Chart for NGC 188 on Sun, September 15, 2024
RA: 0h47m11.52s DEC: 85d14m38.4s



Visibility Chart for NGC 1528 on Sun, September 15, 2024
RA: 4h15m30.72s DEC: 51d13m04.8s

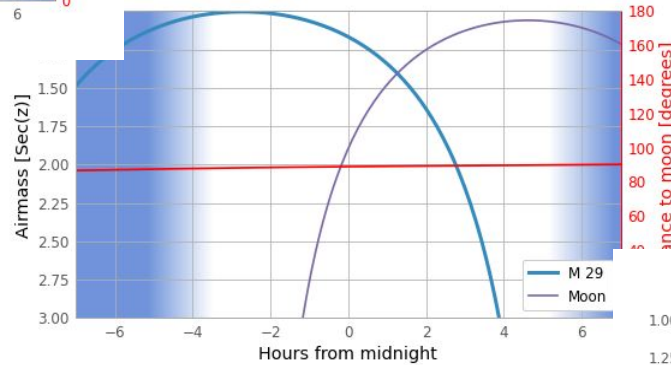


Visibility Chart for M 29 on Sun, September 15, 2024
RA: 20h23m46.32s DEC: 38d29m13.2s

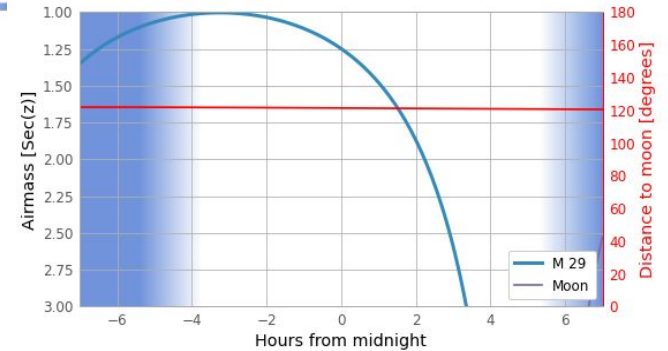


... and the best time for a given target

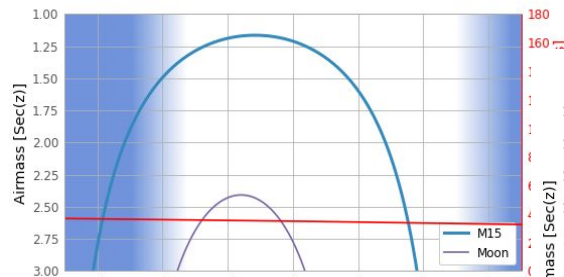
Visibility Chart for M 29 on Sun, September 22, 2024
RA: 20h23m46.32s DEC: 38d29m13.2s



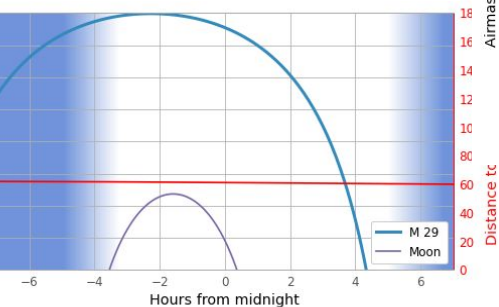
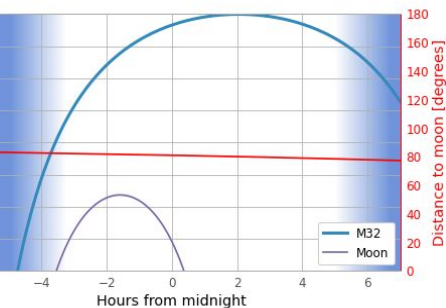
Visibility Chart for M 29 on Mon, September 30, 2024
RA: 20h23m46.32s DEC: 38d29m13.2s



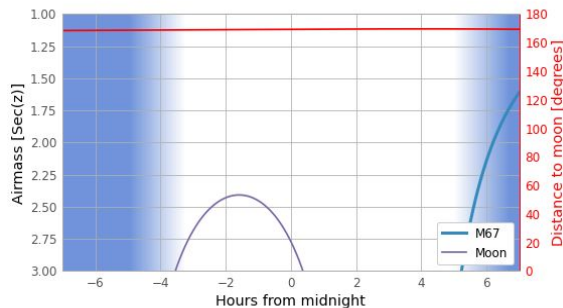
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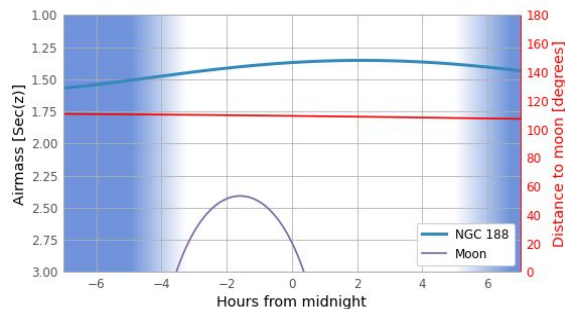
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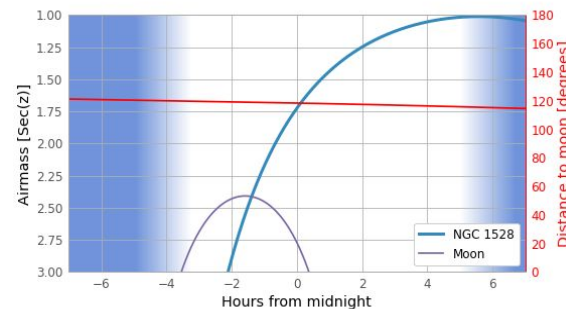
Visibility Chart for M67 on Sun, September 15, 2024
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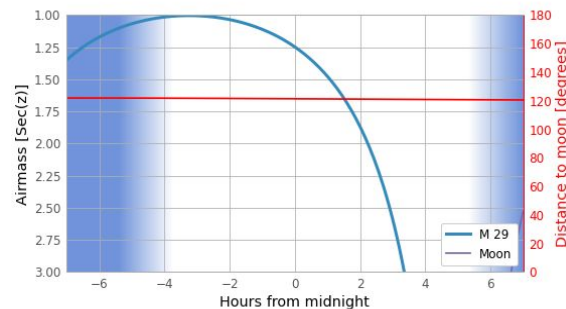
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Visibility Chart for M29 on Mon, September 30, 2024
RA: 20h23m46.32s DEC: 38d29m13.2s



Preparations

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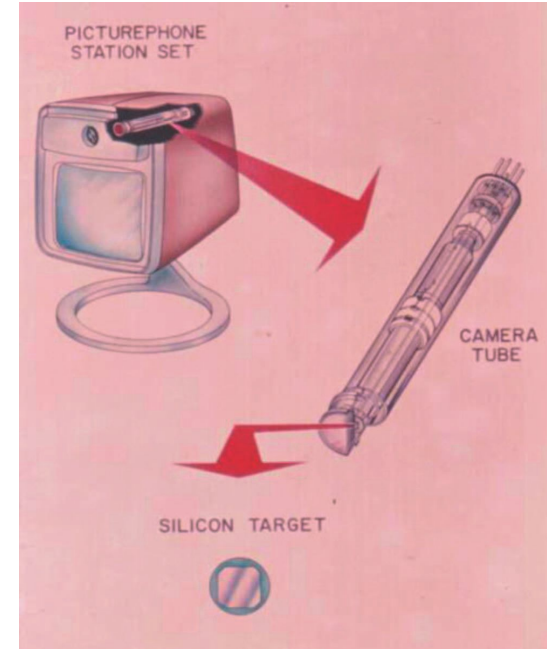
CCDs - the workhorse of UV/Opt/IR astronomers

The Nobel Prize in Physics 2009 was divided, one half awarded to Charles Kuen Kao "for groundbreaking achievements concerning the transmission of light in fibers for optical communication", the other half jointly to Willard S. Boyle and George E. Smith "for the invention of an imaging semiconductor circuit - the CCD sensor"

see Nobel Lecture:

<https://journals.aps.org/rmp/pdf/10.1103/RevModPhys.82.2307>

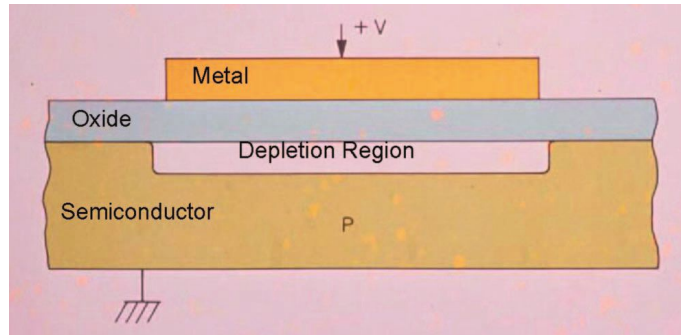
ca. 1969



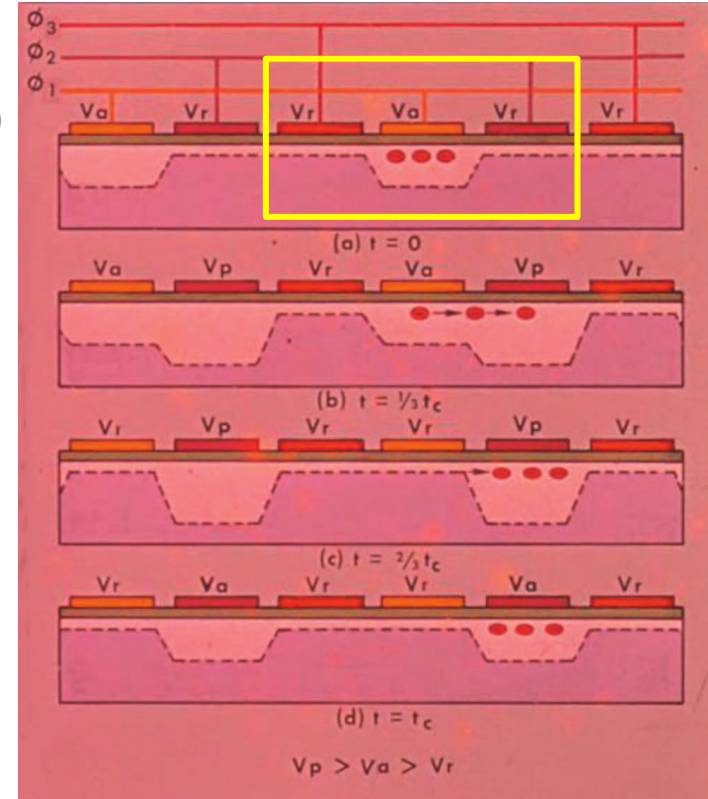
CCDs - the workhorse of UV/Opt/IR astronomers

Electron-hole pairs are created via photo-effect
(similar to solar panels; Einstein's nobel prize 1921)

Charge gets confined & stored by applying electric
voltages



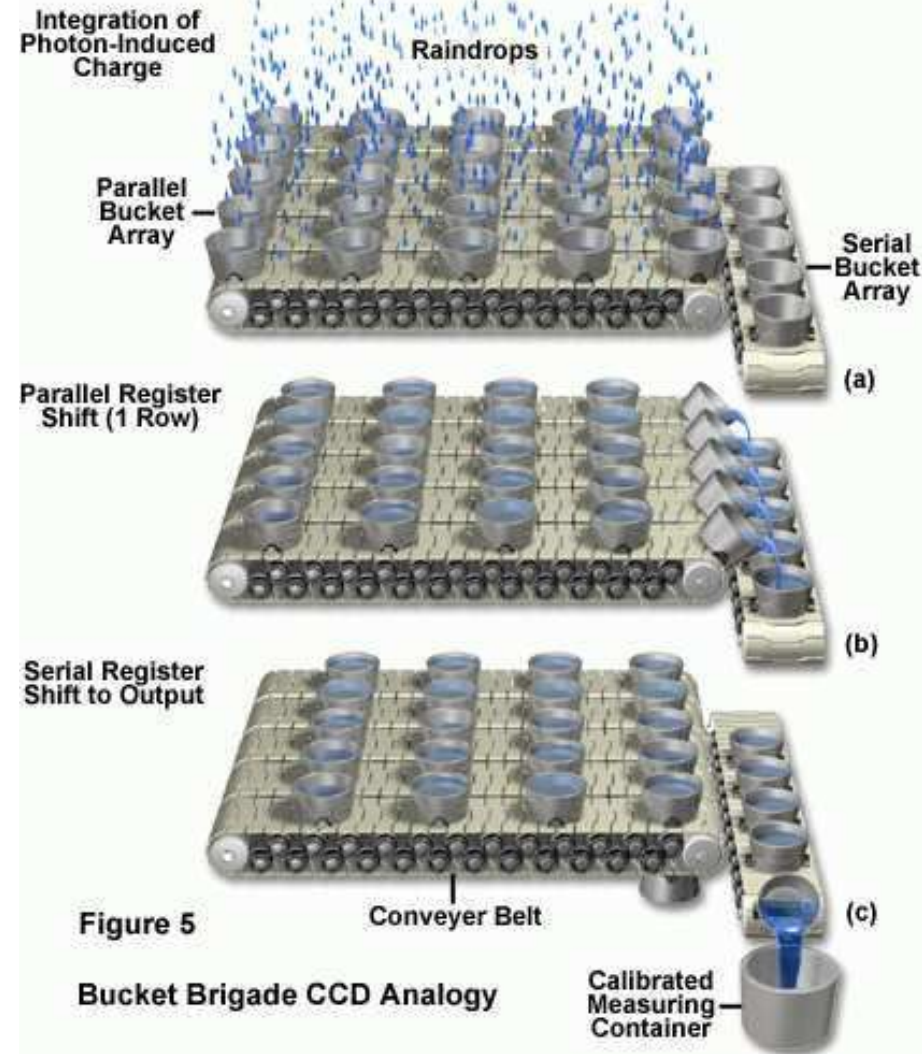
Changing voltages allows to move the charge
→ Need at least three “gates” → tri-gate CCDs



CCDs: Analog detectors

To measure charge in each pixel:

- 1) Move all rows in parallel; every step one row is moved into a special “serial register”
- 2) Then shift only pixels in serial register towards a special pixel connected to an analog-to-digital converter
- 3) Once all pixels in one row have been digitized, move all rows another step
- 4) Repeat until all rows and pixels have been digitized



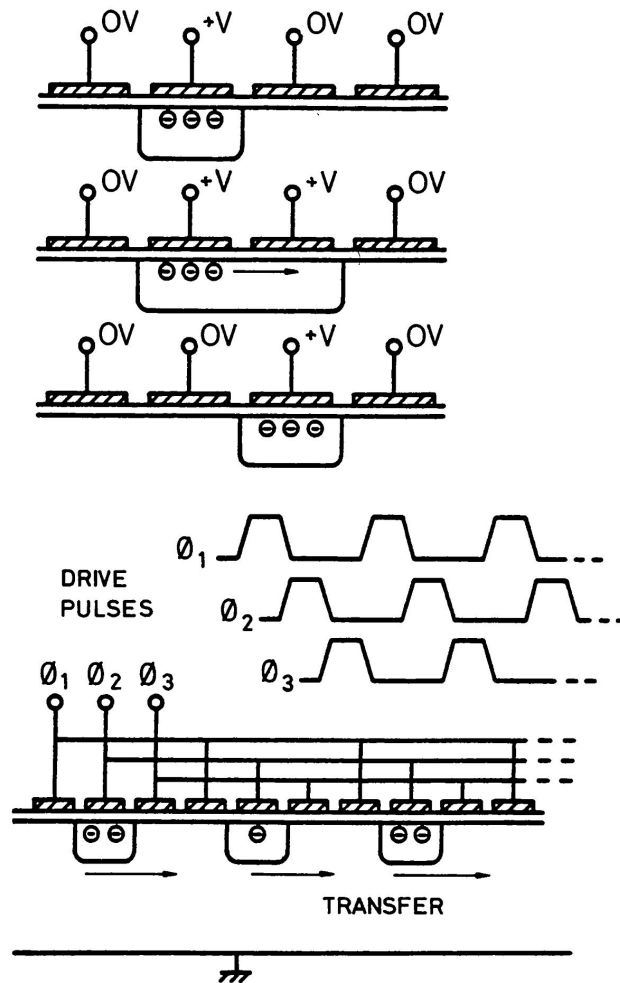


Figure 2a The charge-transfer mechanism employed in three-phase CCDs.

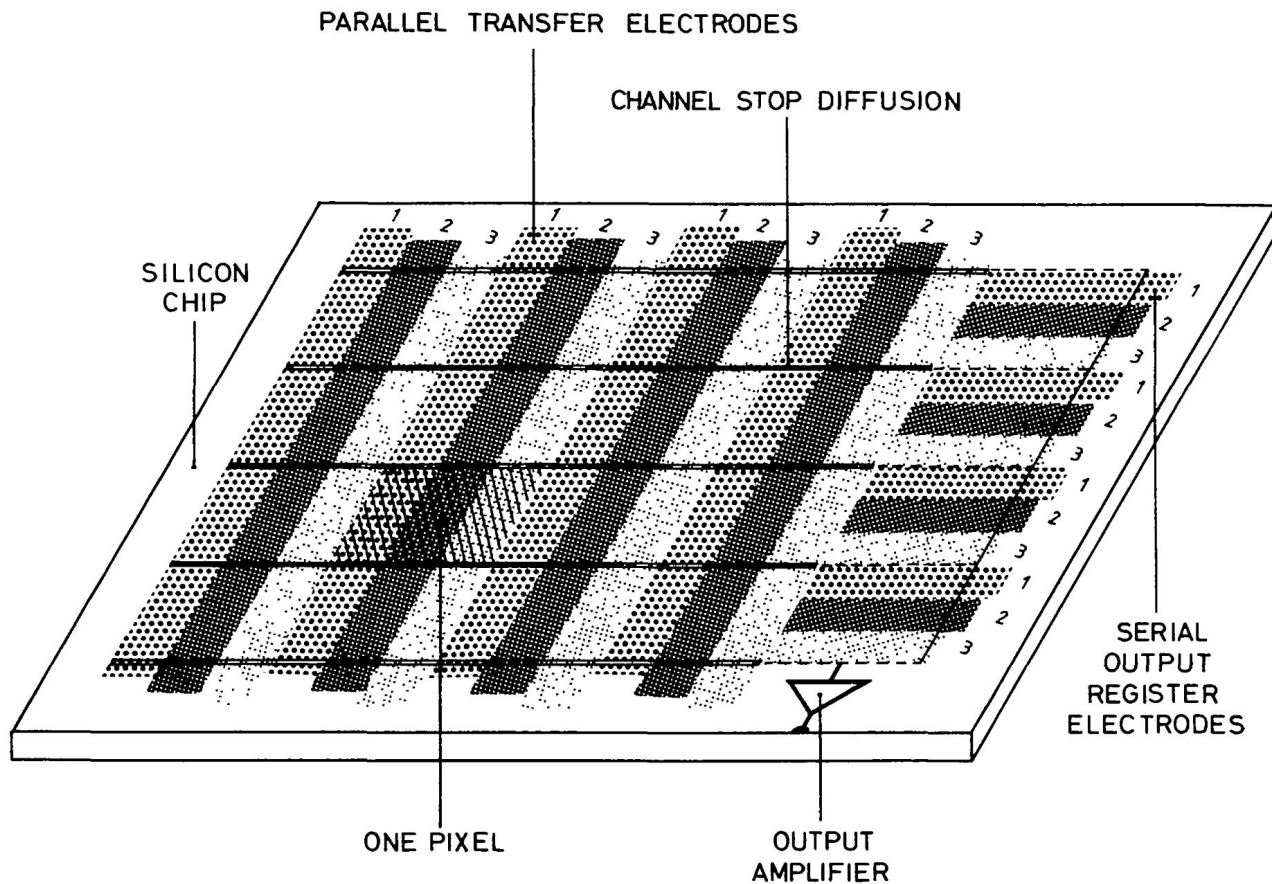


Figure 2b The basic layout of a three-phase two-dimensional CCD. The sequence 1, 2, 3 on each set of electrodes indicates the normal direction of charge transfer in the parallel and serial registers.

CCDs - practical considerations

- Likelihood of generating electron-hole pair from incoming photon depends on wavelength
- Electronic circuitry blocks incoming light, decreasing sensitivity
- Charge transfer efficiency – charge from every pixel needs to be moved lots of times (4K x 4K imager: up to 8000x) ; even small losses quickly add up
- Pixels need to be moved and digitized one by one → slow
- Analog-to-digital converters do not work well at voltages close to zero
- Not all pixels are identical
- Individual defective pixels can have big effects
- Only measuring accumulated charge from any source, not light directly
- Pixels can not hold infinite charge

CCDs - practical considerations

- Likelihood of generating electron-hole pair from incoming photon depends on wavelength
 - Electronic circuitry blocks incoming light, decreasing sensitivity
- Use thinned, back-illuminated CCDs (circuitry is now below the active area)
- For improved red sensitivity make CCDs thicker (more path length for absorption), but: may lose electrons to diffusion → deep-depleted CCDs
- Add microlenses to focus image on photo-active regions
- Add special (organic) coatings to convert UV photons into visible photons

CCDs - practical considerations

- Charge transfer efficiency – charge from every pixel needs to be moved lots of times (4K x 4K imager: up to 8000x) ; even small losses quickly add up
- Pixels need to be moved and digitized one by one → slow

- Not so much a problem anymore nowadays with better manufacturing, but problem remains (e.g. HST cameras)
- Add multiple amplifiers to reduce number of transfers and speed up read-out
- Lower-noise ADC (or electron-multiplying readouts) allow lower noise with faster readout speeds

CCDs - practical considerations

- Analog-to-digital converters do not work well at voltages close to zero
 - Add bias level that needs to be corrected for after readout
- Not all pixels are identical
 - Calibrate using flat-fields
 - critical, but not always trivial

CCDs - practical considerations

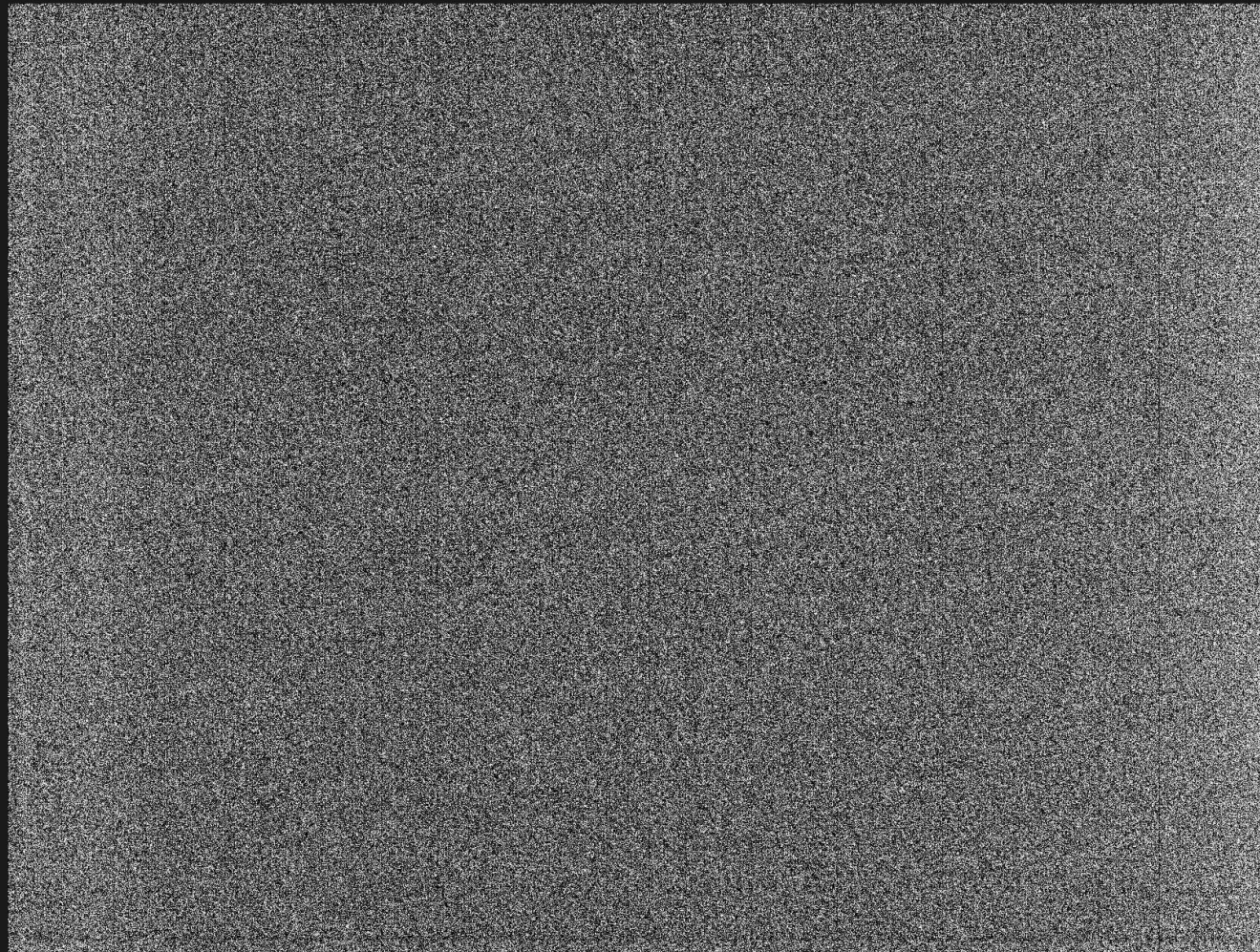
- Individual defective pixels can have big effects
 - minimize by selecting only the best CCDs for science → increased cost
 - special observing techniques to identify and remedy problems in software
- Only measuring accumulated charge from any source, not light directly
 - Obtain dark exposures to statistically remove signal not due to light
 - Cool detectors down to cryogenic temperatures to minimize thermal impacts
- Pixels can not hold infinite charge
 - Choose appropriate detector gain, ADC precision, and exposure time

Bias

single frame:

mostly noise
(which is good!)

no discernable
features
(maybe some
elevated levels
near the edges)



Bias frames

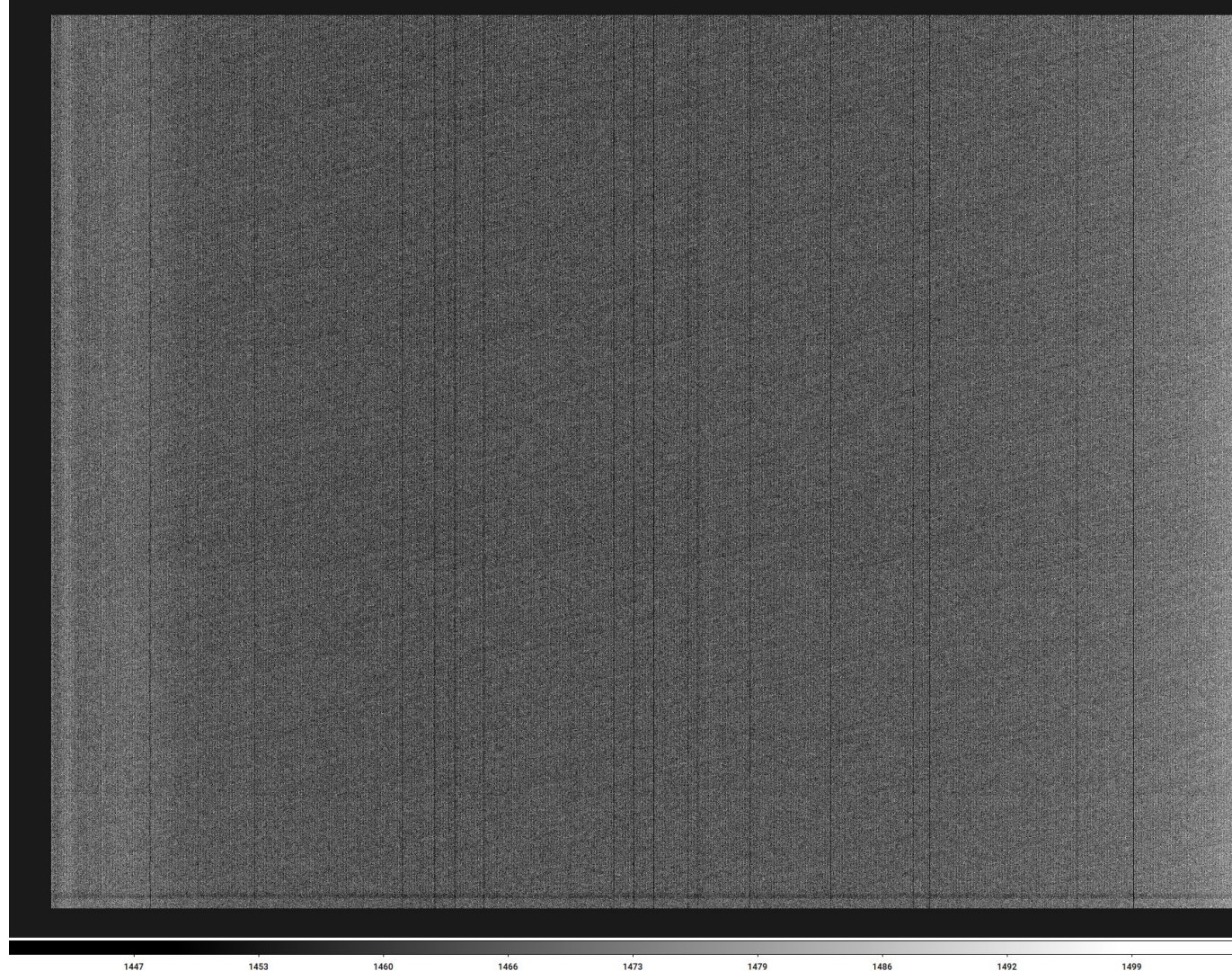
Average of 25
frames:

Elevated levels
near the edges

some banding

several column
defects

horizontal feature
near the bottom



Read-noise and gain

Since CCDs are counting photons, the resulting noise follows a Poissonian distribution:

$$\sigma[e^-] = \sqrt{N[e^-]}$$

However, what we measure from the detector are ADUs, where

$$1 \text{ ADU} = \text{gain} * N[e^-]$$

Therefore, in observational terms:

$$\sigma[\text{ADU}] = \text{gain} * \sqrt{(\text{ADU}/\text{gain})} = \sqrt{\text{ADU}} \times \sqrt{\text{gain}}$$

Read-noise and gain

In practice we also need to consider noise introduced by the readout electronics:
⇒ read noise:

$$\begin{aligned}\sigma^2(\text{total}) &= \sigma^2(\text{read-noise}) + \sigma^2(\text{signal}) \\ &= \sigma^2(\text{read-noise}) + N[e^-]\end{aligned}$$

Estimating read-noise

Read-noise levels (in ADU) can be derived from bias frames that have no signal:

Method 1: Calculate standard deviation across all pixels in a frame

Pro: need only a single frame

Con: Inaccurate if bias has some structure or lots of bad pixels

Method 2: Calculate standard deviation for each pixel across a sequence of bias frames

Pro: Works in all cases

Con: Needs a lot more data

Estimating gain

Earlier we saw: $\sigma^2(\text{total}) = \sigma^2(\text{read-noise}) + \sigma^2(\text{signal})$

In the special case where signal is high enough to neglect the read-noise, we can derive the gain from signal and noise:

$$S_{ADU} = \frac{N[e^-]}{g} \quad \sigma_{ADU} \frac{\sqrt{N[e^-]}}{g} \quad \longrightarrow \quad \sigma_{ADU}^2 = \frac{N[e^-]}{g^2}$$

combining these:

$$\frac{\sigma_{ADU}^2}{S_{ADU}} = \frac{1}{g} \rightarrow g = \frac{S_{ADU}}{\sigma_{ADU}^2}$$

Gain and read-noise (the better way)

In real data, read-noise will never quite be negligible, so we have to take it into account

We can combine bias frames with zero signal with well illuminated data (flat-fields). However, the latter show some large-scale and pixel-to-pixel variations. Using differences between pairs of images eliminates such differences.

Janesick's method (Janesick 2001) allows to estimate gain and readnoise simultaneously, from a pair of biases and flatfields.

Janesick's method

B_1 and B_2 are a pair of bias frames, and F_1 and F_2 are a pair of flat-fields

$$\Delta_B = B_1 - B_2 \qquad \Delta_F = F_1 - F_2$$

$$g = \frac{(\overline{F_1 + F_2}) - (\overline{B_1 + B_2})}{(\sigma_{\Delta_F}^2 - \sigma_{\Delta_B}^2)} \quad [e^- / ADU]$$

$$R = g \times \frac{\sigma_{\Delta_B}}{\sqrt{2}} \quad [e^-]$$

where g and R are gain and read-noise averaged over all pixels in the frame